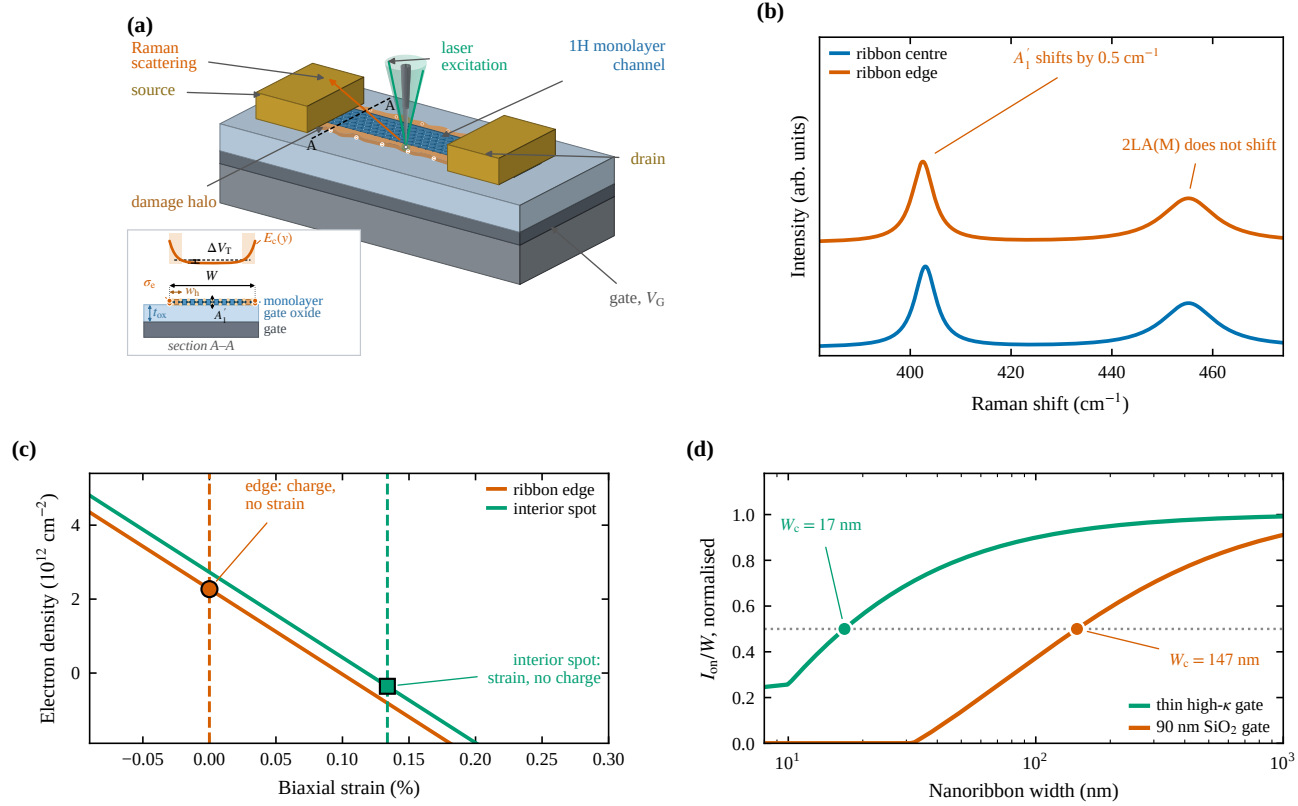

(a)
Raman scattering
source
laser excitation
1H monolayer channel
drain
damage halo
gate, V_G
A
A
ΔV_T
$E_c(y)$
W
σ_e
w_h
monolayer
t_{ox}
A_1'
gate oxide
gate
section A–A
(b)
ribbon centre
ribbon edge
A_1' shifts by 0.5 cm^{-1}
2LA(M) does not shift
Intensity (arb. units)
400
420
440
460
Raman shift (cm^{-1})
(c)
edge: charge, no strain
ribbon edge
interior spot
interior spot: strain, no charge
Electron density (10^{12} cm^{-2})
4
2
0
−0.05
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Biaxial strain (%)
(d)
W_c = 17 nm
W_c = 147 nm
thin high-κ gate
90 nm SiO$_2$ gate
I_{on}/W, normalised
1.0
0.8
0.6
0.4
0.2
0.0
10^1
10^2
10^3
Nanoribbon width (nm)